

Numbers Behind Dominance:

A Data Analysis of Manchester City's 2022-23 Season

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Independent Sport Analytics
Project

Introduction

Manchester City's 2022-23 season is widely considered one of the greatest seasons in modern soccer history, highlighted by their 5 champion titles and consistent dominance through out multiple competitions. Rather than relying on subjective opinions or fan narratives, I want to investigate whether their success could be objectively measured through data. By analyzing key performance metrics such as expected goals, possession, and defensive statistics, I want to use this project to provide a clear, evidence based analysis of what truly set Manchester City apart from other elite teams during that season.

Research Question:

Compared to other Europe's elite clubs in 2022-23 season, what statistical characteristics made Manchester City's season unique?

Motivation:

As both a soccer fan and an enthusiastic statistics/data science student, I want to combine my personal interest with quantitative analysis using data science.



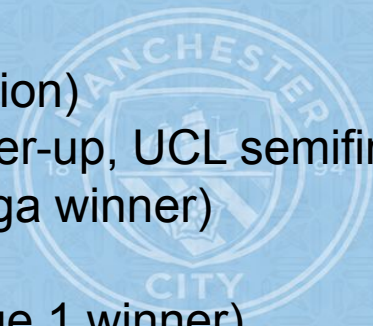
Goals:

- Collect data
- Build a dataset
- Engineer new metrics
- Analyze results using python

Data Construction

Teams:

- Manchester City (Main observation, 5 titles in 22-23 season)
- Arsenal (EPL runner-up)
- Barcelona (La liga champion)
- Real Madrid (La liga runner-up, UCL semifinalist)
- Bayern Munich (Bundesliga winner)
- Napoli (Serie A winner)
- Paris Saint-Germain (Ligue 1 winner)
- Inter Milan (UCL finalist)



Variables

Attack:

- xG (expected goals)
- Big Chances Created
- Shots On Target

Defense:

- xGA (expected goals against)
- Opponent Big Chances Created
- Opponent Shots On Target



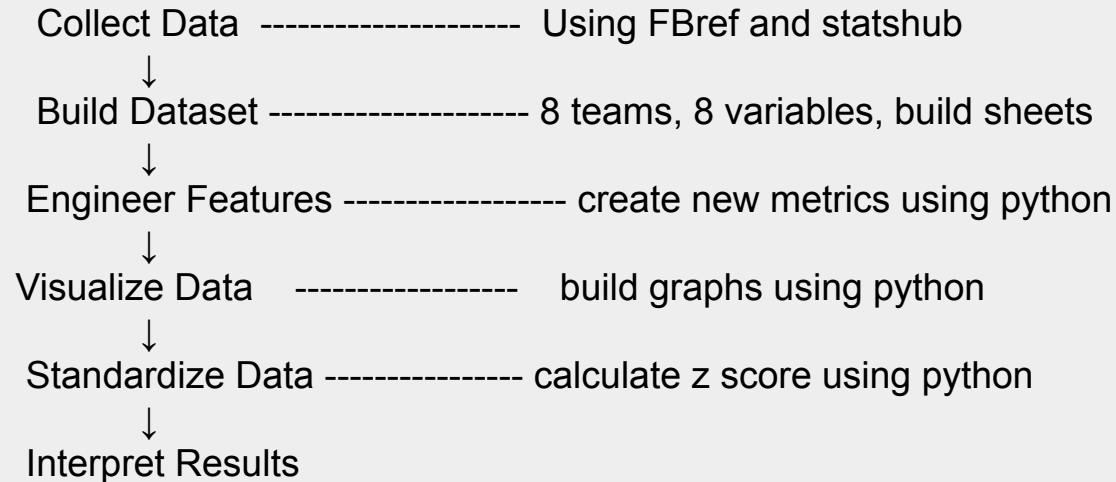
Control:

- Possession
- Passes

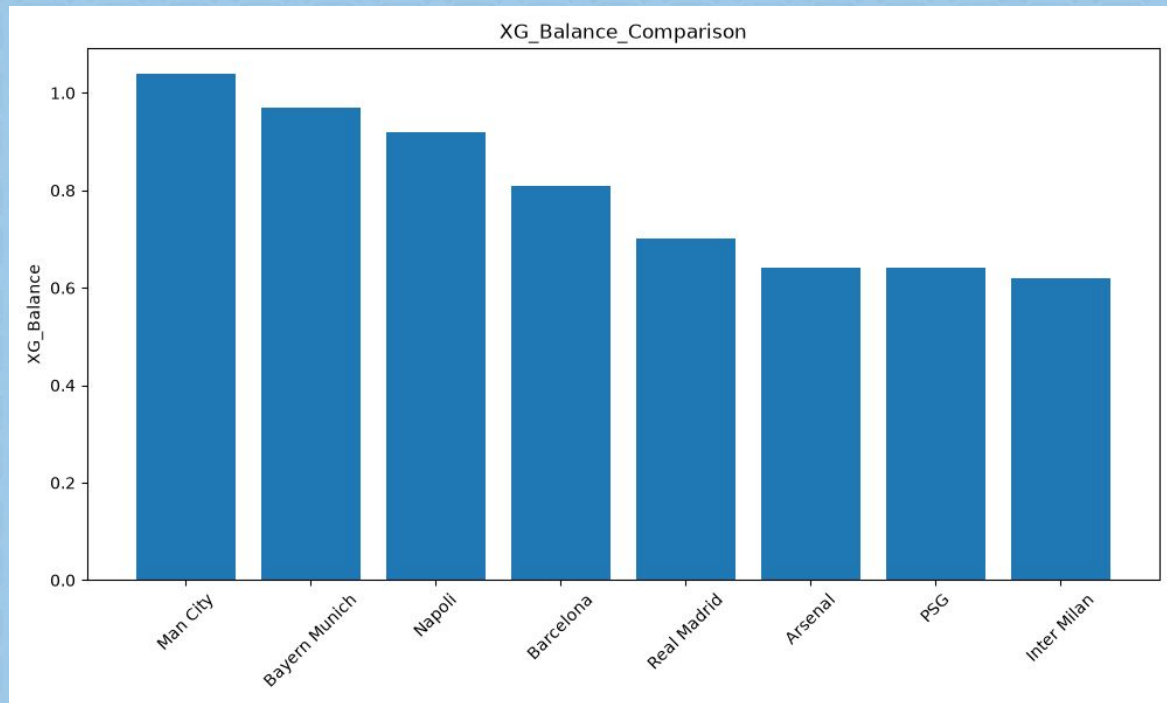
Sources:

- [FBref](#), [statshub](#)

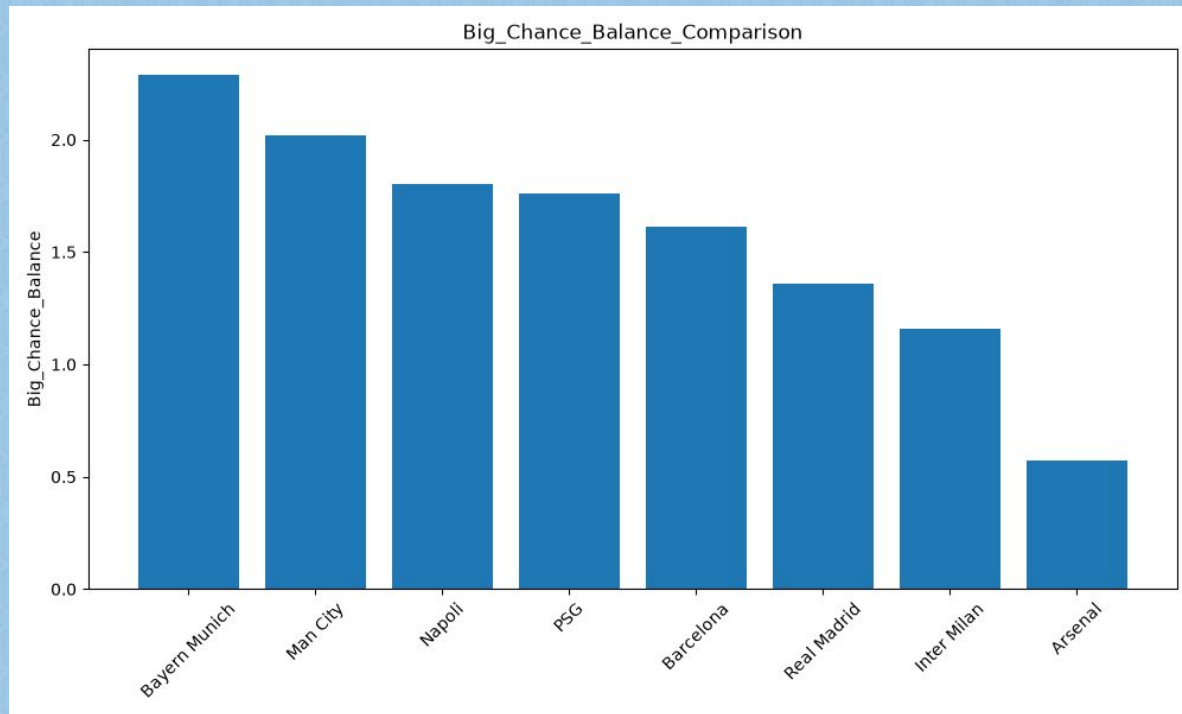
Methodology: From Raw Data to Insights



Two-Way Dominance: Attack and Defense Results



Two-Way Dominance: Attack and Defense Results



Two-Way Dominance: Attack and Defense Results

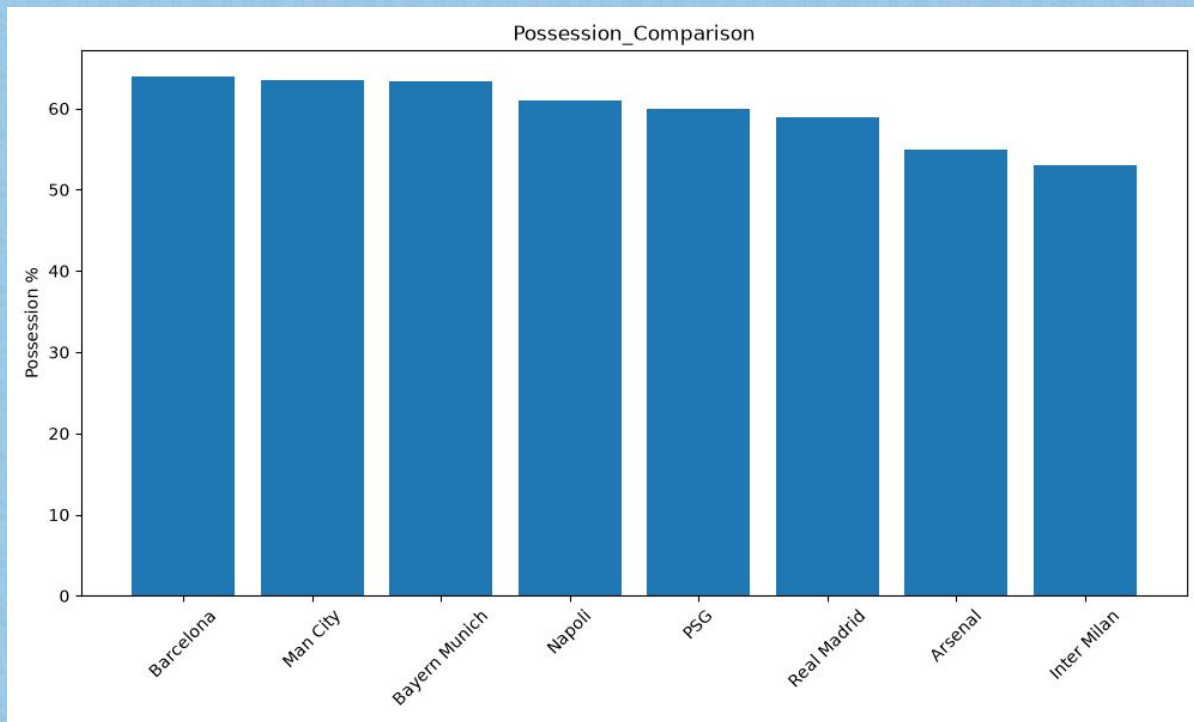
Observation 1:

Manchester City had a **very high xG Balance** compared to other teams, which shows they **consistently created more quality chances** than they allowed. This means that their attacking play was not only effective but also sustained throughout matches.

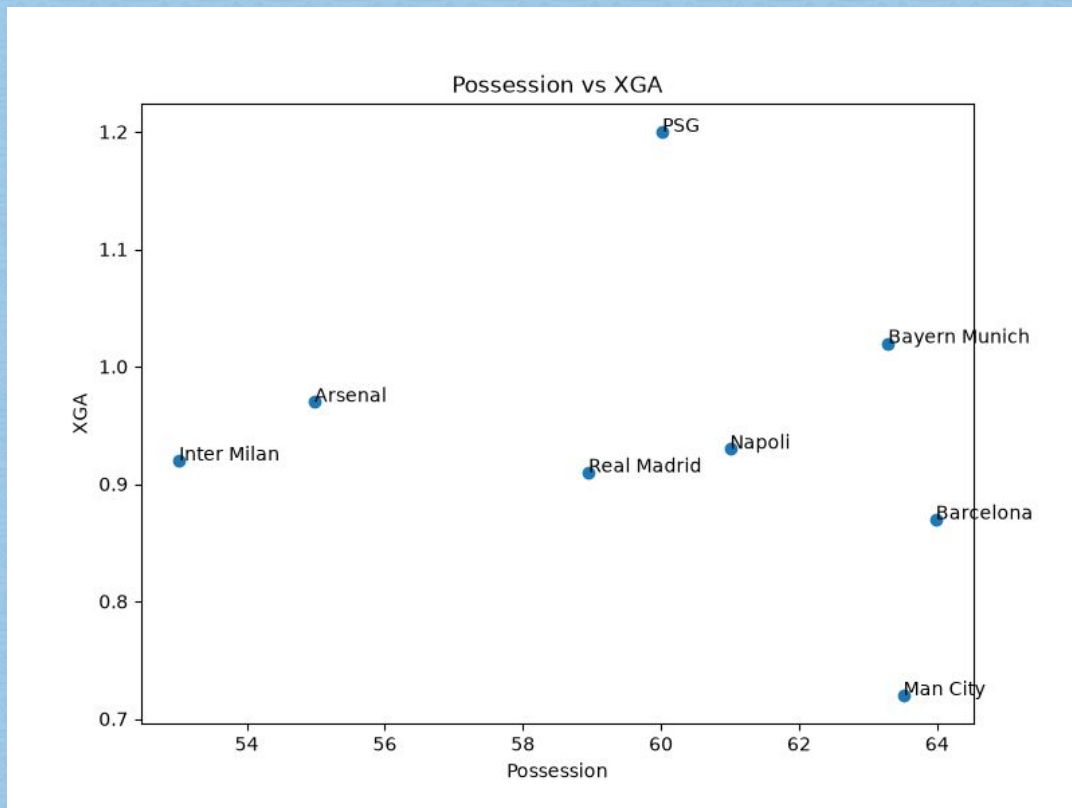
Observation 2:

Their Big Chance Balance was also strongly positive, which means that they **limited opponents' clear scoring opportunities while generating many of their own**. This points to strong defensive organization in addition to attacking strength.

Game Control : Where City Differentiates



Game Control : Where City Differentiates



Game Control : Where City Differentiates

Observation 1:

Although Barcelona, Manchester City, and Bayern Munich all had possession above 63%, their defensive performance differed significantly. Manchester City had the lowest xGA (0.72), Barcelona had a moderate xGA (0.87), and Bayern Munich had a much higher xGA (1.02). ***This suggests that simply controlling the ball was not enough; teams also needed strong defensive organization when possession was lost.***

Observation 2:

Manchester City stood out because ***they paired very high possession (63.51%) with the lowest xGA among all eight teams.*** Unlike PSG and Bayern Munich, who also dominated possession but allowed more expected goals against, City successfully translated ball control into defensive stability. This indicates that ***Manchester City controlled both attacking and defensive sides of the game.***

Which Areas Made City Unique?: Z Score Analysis

What is Z score:

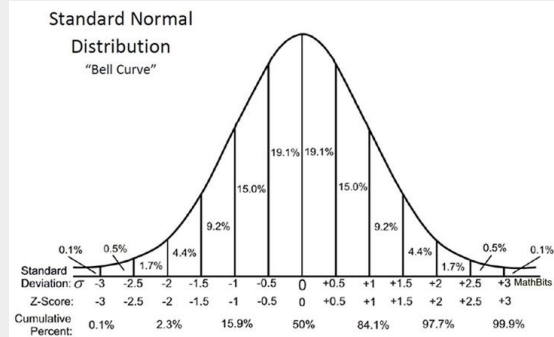
A Z score is a number that tells how far a value is from average, measured in standard deviations. Z scores are used to standardize different performance metrics and compare elite clubs on the same scale. **Positive values** indicate above average performance, **negative values** indicate below average performance, and a **value of zero** represents the average among the selected clubs.



$$z = \frac{x - \mu}{\sigma}$$

μ = Mean

σ = Standard Deviation



Which Areas Made City Unique?: Z Score Analysis

Manchester City 2022-23 Project: Z-Score Analysis

Elite European Clubs Comparison

Team	XG_Balance	Big_Chance_Balance	Shot_Dominance	Possession	Passes
Man City	1.486126	0.834606	0.761553	0.905265	0.884649
Arsenal	-0.915694	-1.862172	-0.94256	-1.20095	-1.571052
Barcelona	0.10508	0.072069	-0.545228	1.021316	0.24088
Real Madrid	-0.555421	-0.392893	-0.359806	-0.220683	0.43889
Bayern Munich	1.065807	1.336765	1.847594	0.850943	0.43589
Napoli	0.76558	0.42544	0.60262	0.285499	0.104926
Inter Milan	-1.035785	-0.764862	-1.127982	-1.68491	-1.52305
PSG	-0.915694	0.351046	-0.236192	0.043519	0.988865

Which Areas Made City Unique?: Z Score Analysis

Observation 1:

Manchester City's greatest advantage was **controlling chance quality**. Among all teams, Manchester City had the **highest xG Balance z-score** (+1.49). This indicates that they **created significantly higher-quality scoring opportunities while simultaneously limiting the quality of chances their opponents created**. Their dominance was built on controlling both attacking and defensive sides rather than simply scoring a large number of goals.

Observation 2:

City combined game control with efficiency better than other elite clubs. Several clubs stood out in individual categories. Bayern Munich had the highest Shot Dominance (+1.85), while Barcelona had the highest Possession (+1.02). However, Manchester **City remained strong across every metric** without any major weaknesses. This balanced profile may explain why they were able to sustain success across multiple competitions and ultimately achieve a historical season.

Conclusion: Answer the Research Question

Research Question:

Compared with Europe's elite clubs in the 2022-23 season, what statistical characteristics made Manchester City's season unique?

Answer:

This analysis conclude that Man City's success cannot be explained by one statistic alone. The team consistently performed above other elite team across multiple dimensions, including chance creation, defensive stability, possession, and passing. This balanced performance distinguished Man City from other top European clubs.

Conclusion: Summarize Key Findings

Finding 1: Superior Chance Control

Manchester City has the highest xG Balance among all teams analyzed. This indicates that they consistently created higher quality and scoring opportunities while limiting the quality of chances available to their opponents.

Finding 3: Possession Is Not Enough

The possession analysis showed that simply controlling the ball does not guarantee defensive success. Teams such as Barcelona also have high possession but allowed considerably more expected goals against than Manchester City. This shows that effective possession depends on how well it is translated into defensive organization and game control.



Finding 2: Balanced Rather Than Specialized

Several clubs surpassed City in individual areas. For example, Bayern Munich achieved the highest Shot Dominance, while Barcelona recorded the highest average possession. However, Manchester City maintained above average performance across every field without displaying a significant weakness. This balanced performance contributed to their ability to compete successfully in multiple competitions throughout the season.